

A Computational Comparison of Prototype, Exemplar, and Rule-Based Models of Human Category Learning

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Abstract—I propose a small computational tool that implements three competing models of how a concept is represented in the mind—a prototype model, an exemplar model (the Generalized Context Model), and a rule-based model—and evaluates them on the classic Shepard, Hovland, and Jenkins category structures. The question I want to answer is not which model classifies most accurately, but which one generalizes the way people do, reproducing the human difficulty ordering across the six structures. Treating each algorithm as a theory of mind rather than a classifier keeps the project grounded in cognitive science, and the well-defined behavioral benchmark makes the comparison falsifiable. I outline the research question, its cognitive science framing, the system design, expected outcomes, and a phased plan scoped to roughly seventy hours.

I. TOPIC AND RESEARCH QUESTION

A child who has seen three or four dogs can pick out the next dog without ever being handed a definition, and a machine classifier usually cannot, and the reason I keep coming back to is that the gap is less about how much data each one sees than about what a concept is and how it gets stored. Cognitive science gives three durable answers, which I take as my starting point: a concept might be an average of its members, the prototype view, or the full set of remembered instances, the exemplar view, or an explicit definition stated as conditions, the rule view. Each of these predicts a different pattern of generalization. That is the point I want to build on, because if the three views generalize differently then they can be told apart, and a question about them can be made to have a wrong answer.

My research question is narrow and I have stated it so that it can fail: among prototype, exemplar, and rule based representations, which one best reproduces the way people generalize categories learned from a handful of examples, measured against the human difficulty ordering of the six Shepard, Hovland, and Jenkins (SHJ) category structures [1]? I am drawn to this because of my earlier work in this course, where I worked a single problem through competing forms of representation and found that the choice of representation, not the cleverness of the search on top of it, decided whether the account held up, and I want to put that intuition under a

test where the three representations are forced to make visibly different predictions rather than left to agree.

The question matters because it sits on the working boundary between human cognition and artificial intelligence. People still generalize from one or a few examples in ways current systems struggle to match [2], and recent work comparing language model embeddings against human categorization data finds the machine representations line up with human category boundaries only at a coarse level and come apart on the finer structure of a concept [3]. So asking which classical representation best matches human generalization is not a historical exercise. It is a question about what a more human aligned representation would have to look like.

II. COGNITIVE SCIENCE FRAMING

This is a project about mental representation, which is the central concern of the computational and representational understanding of mind that frames the course [4], and so the three models are not three classifiers that happen to wear cognitive labels but three claims about what is in the head when a person holds a concept, each descended from its own tradition. The prototype account comes out of Rosch's work on typicality and graded membership [5] and Posner and Keele's finding that people abstract a central tendency they were never actually shown [6]. The exemplar account, set out in Medin and Schaffer's context theory [7] and made precise in Nosofsky's Generalized Context Model [8], holds that no average is stored at all and that classifying something is just judging its similarity to the instances already in memory. The rule account treats a concept as necessary and sufficient conditions, and it is the view the other two were built to replace.

What makes each of these a claim about how minds work, rather than a choice of algorithm, is that each one predicts different behavior. A prototype model says typical items should be judged faster and more confidently than odd ones. An exemplar model says specific past cases should matter and exceptions should be handled gracefully. A rule model says boundaries should be sharp and fuzzy categories should be hard.

The Shepard, Hovland, and Jenkins structures are what turn those predictions into a test I can lose, because people learn the six types in a stable order of difficulty, and that order is the behavioral signature any honest model of human concepts has to reproduce [1]. I treat reproducing that ordering as necessary: a model that classifies accurately but gets the ordering wrong has failed the cognitive claim, whatever its accuracy. Raw accuracy I treat as helpful but not sufficient, a thing to report rather than the thing to win on. These ideas come from the course readings on representation and concept types [9] and from the categorization literature those readings point into.

III. INITIAL SYSTEM DESIGN AND EXPECTED OUTCOMES

What I will build. The tool is a small Python codebase with three parts: an encoding of the SHJ stimuli (eight items over three binary features, split into the six types), the three models, and a runner that trains and tests each under matched conditions and produces the comparison figures. A reproducible notebook is the core deliverable and demonstration; a lightweight interactive front end is a possible extension for the presentation, not part of the core.

How the cognitive science shapes the design. Each theory from Section II maps to one model. The prototype model stores one summary point per category and classifies by similarity to it [5], [6]. The exemplar model stores every training item and classifies a probe by summed, attention-weighted similarity to those instances, which is the GCM directly [7], [8]. The rule model induces an explicit feature-based boundary. The design is driven by theory, not by engineering convenience: the differences among the models are the differences among the theories.

What I expect to learn. I expect the rule model to win on the simplest single-feature structure and to fail on the type people find unexpectedly easy, while the exemplar model should better track the overall human ordering [8]. The headline finding is which model's errors match the human difficulty order, not which scores highest. A secondary check on a smaller structure that historically separates prototype from exemplar accounts [7] tests whether the exemplar advantage is specific or general. If no model reproduces the ordering, that is an informative negative result, and I will report it as one.

IV. ETHICS AND CITI TRAINING

This project uses published, fully specified artificial category structures and synthetic stimuli. It involves no human participants and no collection of personal data, so CITI training is not applicable.

V. PROJECT PLAN AND TIMELINE

The phased plan totaling approximately seventy hours appears in the Appendix.

REFERENCES

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APPENDIX

APPENDIX: PROJECT PLAN AND TIMELINE

The plan below follows the provided course planning template. Total estimated effort is approximately 70 hours. Milestone deadlines are shown as divider rows.

Week	Task	Task Description	Est. Hrs	Done
2	1	Create task list and choose research question	0.5	Y
2	2	Survey concept-learning literature and confirm tool track / scope	3.5	N
3	3	Collect and verify core scholarly references	1.5	N
3	4	Write project pitch (all sections) and format in IEEE L ^A T _E X	3.5	N
PROJECT PITCH DUE				
4	5	Set up repository, environment, and project structure	1.5	N
4	6	Encode SHJ category structures and stimuli	3	N
4	7	Implement the three core models (prototype, GCM, rule-based)	11.5	N
5	8	Build experiment runner and run models on SHJ structures	5	N
5	9	Generate comparison figures and draft optional check-in	4	N
OPTIONAL MIDPOINT CHECK-IN DUE				
6	10	Code debugging and architecture refactoring	7	N
6	11	Add Medin & Schaffer 5-4 secondary check	2.5	N
6	12	Analyze model fit against human difficulty ordering	3	N
6	13	Expand reference set to 10+ for final report	1.5	N
7	14	Draft full report (abstract through conclusion)	10	N
8	15	Revise report, integrate citations, finalize repository	6	N
8	16	Update project plan and submit final report	1.5	N
FINAL REPORT DUE				
9	17	Build poster and write/rehearse 5-minute script	4	N
9	18	Record video, assemble submission PDF, verify and submit	3.5	N
FINAL PRESENTATION DUE				
Total Estimated Hours			73	

Statement on AI use. Generative AI (Claude) was used for IEEE/L^AT_EX formatting, phrasing refinement, structural organization, and verification of reference details. The research question, cognitive science framing, model selection rationale, and interpretation are my own.